

Review Article

Role of Ultrasound-Based Therapies in Cardiovascular Diseases



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ABSTRACT

Cardiovascular diseases (CVDs) remain the leading cause of morbidity and mortality globally, placing an immense burden on health care costs worldwide. The emergence of therapeutic ultrasound-based therapies in the CVD management represents a promising innovative strategy beyond current established approaches. This paper explores three distinct modalities of ultrasound-based therapies—high-intensity focused ultrasound, extracorporeal shock wave therapy, and low-intensity pulsed ultrasound—each characterized by unique acoustic parameters and mechanisms of action tailored to specific therapeutic outcomes. High-intensity focused ultrasound was shown to be beneficial as an adjunct in the treatment of myocardial infarction and arrhythmias. It has also been investigated for the *in vivo* treatment of resistant hypertension, symptomatic aortic valve stenosis, arterial stenosis, tumors, hypertrophic cardiomyopathy, and external cardiac pacing. Extracorporeal shock wave therapy was shown to be beneficial in the treatment of chronic refractory angina pectoris, while low-intensity pulsed ultrasound was shown to be beneficial in dissolving blood clots and improving blood flow in the treatment of acute pulmonary embolism, despite its association with an increased risk of bleeding. Ultrasound-based therapies are, therefore, a potential adjunct and comparatively safe adjuncts for managing challenging CVD cases. Further investigations are essential to validate their long-term effectiveness and safety, particularly for high-risk individuals susceptible to postprocedural complications.

ABBREVIATIONS

AV, aortic valve; CVD, cardiovascular disease; ECSWT, extracorporeal cardiac shock wave therapy; ESWT, extracorporeal shock wave therapy; HIFU, high-intensity focused ultrasound; LIPUS, low-intensity pulsed ultrasound.

Introduction

Cardiovascular diseases (CVDs) continue to be the world's leading causes of morbidity and mortality, with a major impact on health care expenditure.^{1,2} Diagnostic medical ultrasound with frequencies in the megahertz range (MHz), including Doppler ultrasonography, has established clinical utility in the diagnosis of cardiovascular structural and functional abnormalities, including intracardiac and blood flow hemodynamics within the heart and blood vessels. The physical properties of

ultrasound, which are acoustic energy waves with mechanical and thermal effects on insonated biological tissues, can be harnessed for diagnostic medical imaging as well as therapeutic purposes. The acoustic energy of ultrasound consists of mechanical longitudinal waves with pressure amplitudes that can be spatially directed and focused as a geometric beam with acoustic power and intensity, is measured in joules or watts per square centimeter (W/cm²). Ultrasound can be transmitted as intermittent pulses (pulsed-wave ultrasound) or continuously (continuous-wave ultrasound). During transmission and interaction with

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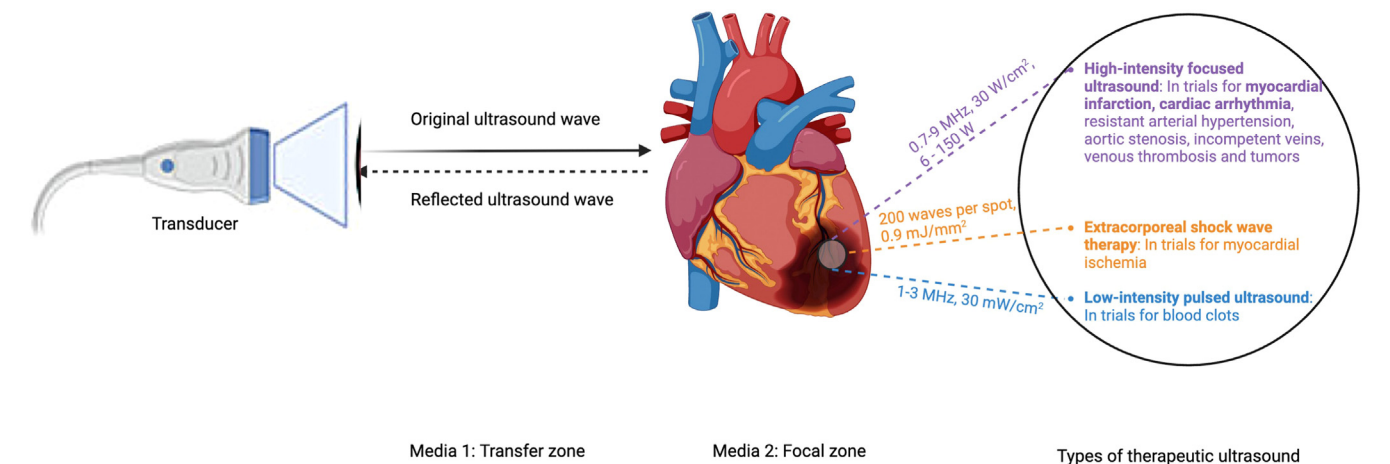


Figure 1. Cardiovascular ultrasound-based therapies. Created with BioRender.com (2023).

tissues, portions of the acoustic energy within the ultrasound beam are reflected, transmitted, and absorbed, depending on the acoustic impedance of the insonated tissues. The extent of these interactions is also influenced by the ultrasound instrumentation and technique, including operating mode, transducer design, application technique, and the total acoustic energy output.^{3,4}

For several years, researchers have investigated the therapeutic use of ultrasound. More recently, there has been a surge of interest in therapeutic ultrasound for the treatment of CVDs. Advances in bioengineering and endovascular treatments have allowed for more accurate delivery of high-powered ultrasound, opening new avenues for the therapeutic use of ultrasound^{5–8} (Figure 1). The therapeutic benefits of ultrasound are determined by its frequency, power, and whether it is targeted. This paper presents a contemporary overview of therapeutic applications of ultrasound in patients with CVDs.

Ultrasound-Based Therapies: Mechanisms of Action in CVDs

Ultrasound-based therapies are commonly associated with physical therapy. In the realm of CVDs, ultrasound is seldom used therapeutically and is still under investigation. Its mechanism of action is further summarized in Tables 1 and 2, Figure 1, and below, based on the type of sound waves.

High-Intensity Focused Ultrasound

High-intensity focused ultrasound (HIFU) is a noninvasive therapeutic technique that delivers high-powered, focused ultrasound energy to targeted tissues. In 1942, Lynn first introduced HIFU for focal tissue ablation with minimal/no effects on surrounding tissue.¹³ The ultrasound parameters of HIFU include continuous waves with generally low

frequencies (frequency: 0.7-9 MHz) and high intensities (30 W/cm² and acoustic power: 6-150 W).^{3,6,14–17} Lower frequencies enable deeper penetration for transthoracic approaches with less tissue attenuation and low thermal index¹⁸, whereas higher frequencies (3.8-9 MHz) are suited for more superficial structures due to lower depth of penetration caused by greater attenuation.^{14–17} Ultrasound at frequencies in the MHz range can be safe and effective for treating ischemic stroke, as reported in 57 small clinical trials and in a meta-analysis of 10 studies.^{12,19} The biological effects of HIFU are generated in situ based on the properties of tissues and mediated by thermal and/or mechanical effects.²⁰ HIFU’s thermal effects can be harnessed in cardiovascular medicine to ablate and isolate focal tissues that trigger atrial fibrillation, as well as in noncardiac applications such as renal denervation and malignancies.²⁰ Khokhlova et al. (2014) investigated HIFU for noninvasive cardiac ablation in a porcine model, demonstrating its potential for precise cardiovascular tissue interventions not often associated with ultrasound techniques.²¹ In the “boiling histotripsy” method, HIFU mechanically generates mm-sized boiling bubble that interact with the ultrasonic field to fractionate porcine tissue into subcellular debris without producing additional thermal effects.²¹ Pulsed cavitational ultrasound therapy (thrombotripsy) has also been investigated for the recanalization of proximal deep venous thrombosis in a swine model.¹¹

Extracorporeal Shock Wave Therapy (ESWT)

Extracorporeal shock wave therapy (ESWT) techniques employ high-pressure, high-energy, short duration pulses of sound called shock waves generated outside the body (extracorporeally) to deliver energy to tissues. ESWT produces a mechanical action in the tissue, resulting in rapid pressure changes, tissue deformation, and cellular mechano-transduction. As shockwaves propagate through body tissues, it is the properties of the

Table 1
Thermal vs. nonthermal effects of ultrasound therapy

Therapy	Type	Mechanism	Applications
High-intensity focused ultrasound (HIFU)	Thermal	High-energy focused ultrasound waves induce tissue coagulation and ablation, concentrating energy to elevate temperature precisely in targeted areas, sparing surrounding tissue. ³	- Ablation of arrhythmogenic tissue in atrial fibrillation - Noninvasive renal denervation for resistant hypertension
Extracorporeal shock wave therapy (ESWT)	Nonthermal	Utilizes mechanical effects of ultrasound waves such as vibration and pressure to stimulate biological responses without significant heat generation. ^{6,9}	- Treatment of myocardial ischemia by breaking up arterial plaques - Refractory angina pectoris
Low-intensity pulsed ultrasound (LIPUS)	Nonthermal	Low-energy pulsed ultrasound waves activate cellular processes and promote microbubble cavitation, facilitating healing and perfusion enhancement. ¹⁰	- Enhancing muscle perfusion - Stroke recovery
Pulsed cavitational ultrasound therapy	Nonthermal	Ultrasound pulses create cavitation bubbles that mechanically disrupt tissues on a microscale for therapeutic benefits. ¹¹	- Cardiac dysfunction amelioration - Noninvasive cardiac ablation - Enhancing drug delivery efficiency

Table 2

Invasive vs. noninvasive ultrasound therapy

Therapy	Type	Method	Applications	Specific considerations
Extracorporeal shock wave therapy (ESWT)	Noninvasive	Therapy applied externally without the need for surgical intervention. ^{6,9}	- Myocardial ischemia - Refractory angina pectoris	- Patient comfort - Repeat sessions may be required
Pulsed cavitational ultrasound therapy	Noninvasive	Uses external devices to deliver therapeutic ultrasound waves. ¹¹	- Aortic valve stenosis - Noninvasive cardiac ablation	- Focused ultrasound to specific areas without surgery
Low-intensity pulsed ultrasound (LIPUS)	Noninvasive	Ultrasound therapy is delivered through the skin using portable or fixed devices. ¹⁰	- Stroke recovery - Cardiac dysfunction - Home or clinic-based therapy for chronic conditions	- Regular sessions needed for effect
High-intensity focused ultrasound (HIFU)	Invasive	Therapy requiring surgical exposure or insertion into the body. ³	- Ablation of arrhythmogenic tissue in AFIB during surgeries	- Surgical risks involved - Requires anesthesia
HIFU (invasive approaches)	Invasive	Invasive applications during interventional procedures. ¹²	- Renal denervation	- Direct tissue targeting - Used in resistant hypertension cases

Abbreviations: AFib, atrial fibrillation; HIFU, high-intensity focused ultrasound.

shockwave, e.g., its pressure, focus, and frequency that determine the depth of penetration and interaction with the target tissue. Best practice ESWT clinical treatment parameters and protocols are generally categorized into three energy categories based on total energy dose and the energy per unit area or energy flux density using the following ranges: low (<0.08-10 mJ/mm²), medium (0.08-0.28 mJ/mm²), and high (>0.29-0.60 mJ/mm²). The ESWT protocols include exposure to separate pulses with high amplitude and predominant nonthermal biomechanical effects (frequency: individual shock waves; acoustic power: 0.9 mJ/mm²; and delivery protocol: 200 shock waves per spot, 40 to 60 spots per session, three sessions per week for 3 months).^{6,7}

Low-Intensity Pulsed Ultrasound (LIPUS)

Low-intensity pulsed ultrasound (LIPUS) is a therapeutic modality that harnesses low-power, low-intensity ultrasound (intensities typically around 30 mW/cm², frequencies 1 to 3 MHz, pulse duration of 200 μ s, and pulse repetition frequency of 1 kHz)²² to exert a nonthermal biological effects on tissues, including reduced inflammation, improved blood flow, and healing by direct stimulation of tissues or indirectly by harnessing the therapeutic potential of lipid-coated or drug-loaded microbubbles.^{7,10,23} Microbubbles are microscopic, highly reflective gas-filled bubbles encapsulated in a stabilizing protein, lipid, or polymer shell that oscillate and cavitate when exposed to the mechanical pressures from ultrasound insonation.

Clinical Applications of Ultrasound-Based Therapy in CVDs

Cardiovascular ultrasound-based therapies are a fast-developing discipline with the potential to become useful adjuncts in the treatment of CVDs. Therapeutic ultrasound may be utilized alone, as in stroke (sonolysis) to cause clot lysis and then recanalization,²⁴ or combined with tissue-type plasminogen activator treatment to increase its effects (sonothrombolysis), or in conjunction with microbubbles.^{12,25} The following discourse and Table 2 summarize various clinical applications of therapeutic ultrasound modalities based on the mode of delivery, which includes invasive and noninvasive techniques.

Noninvasive Techniques

Sonothrombolysis

Sonothrombolysis holds significant promise due to its potential to eliminate or significantly reduce reliance on thrombolytic agents for clot dissolution. The primary mechanism of action of sonothrombolysis is mediated through acoustic radiation force, localized tissue displacement, and clot deformation.²⁶ Animal studies showed that LIPUS could be beneficial in **stroke**,²⁷ **ischemic heart disease**,²⁸ and contractile

dysfunction,²⁹ likely through upregulation of vascular endothelial growth factor (VEGF) and endothelial nitric oxide synthase.²⁸ The ultrasonic vibrations induce cells to release proteins and other chemicals that aid in healing and inflammation reduction. A study³⁰ found that lower-frequency ultrasound is more effective at dissolving blood clots than higher-frequency ultrasound. This finding is important because it suggests that ultrasound-induced blood clot dissolution may be a viable alternative to thrombolytic drugs in certain settings.

Sonothrombolysis in Myocardial Infarction. Sonothrombolysis is a therapeutic intervention that combines ultrasound techniques combined with thrombolytic therapy to enhance the treatment of myocardial infarction, particularly ST-segment-elevation myocardial infarction. It is conducted by a trained medical professional and lasts for approximately 20 minutes or until the patient reaches the hospital or interventional laboratory. The procedure involves the use of proprietary microbubble infusions with simultaneous transthoracic echocardiography. High mechanical index (MI) instrument settings (MI of approximately 1.0 to 1.2) with transmit frequency of 1.6 MHz and pulse duration less than 5 μ s with imaging averaging 20 frames³¹ (Figure 2A). Typically, 30 to 60 impulses are administered, with 4 to 8 seconds between each high MI pulse to ensure sufficient replenishment of the microbubble contrast within the perfusion beds.

Preprimary percutaneous coronary intervention (pPCI) sonothrombolysis was shown to be an important factor in improving myocardial salvage in ST-segment-elevation myocardial infarction in the REDUCE Pilot trial.³⁸ It improved myocardial salvage by an additional 17% compared to post-pPCI sonothrombolysis alone. This benefit was not affected by the duration of post-pPCI sonothrombolysis.³⁸

Sonothrombolysis in Cerebrovascular Stroke. After accumulating a substantial body of preclinical evidence regarding the potential efficacy of sonothrombolysis in addressing middle cerebral artery occlusions, numerous clinical trials have been conducted. These trials have employed transcranial Doppler ultrasound for treating middle cerebral artery occlusions, both with thrombolytic agents alone³⁹⁻⁴¹ and in combination with microbubbles^{32,33,42} (Figure 2B).

Extracorporeal Cardiac Shock Wave Therapy

Research has shown that a low level of extracorporeal cardiac shock wave therapy (ECSWT) can upregulate VEGF, a potent mitogen that triggers angiogenesis in human umbilical vascular endothelial cells to ameliorate myocardial ischemia and refractory angina pectoris.^{3,43-46} Cardiac shockwave treatment includes three times a week, 200 shoots/spot at 0.09 mJ/mm² for 45 spots in the ischemic area each time

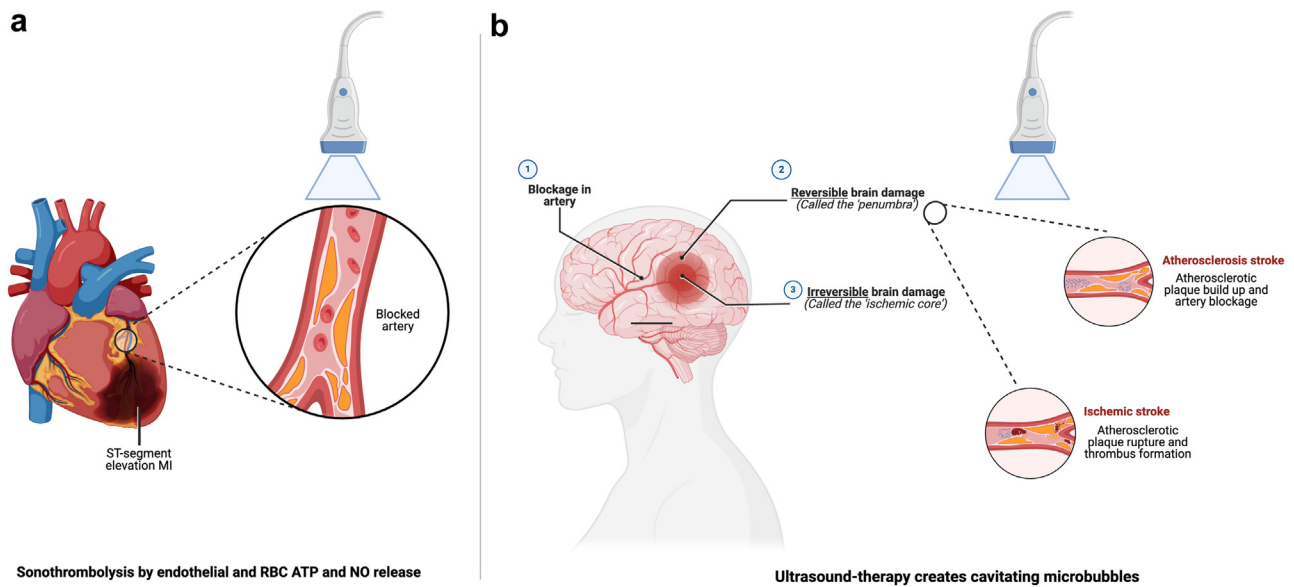


Figure 2. Sonothrombolysis in cardiovascular diseases. Adapted from “Common Types of Stroke,” “Reversible vs. Irreversible Brain Damage Following an Ischemic Stroke,” and “Myocardial Infarction (Heart Attack)” by BioRender.com (2024). References^{32–37}
Abbreviations: ATP, adenosine triphosphate; MI, myocardial infarction; NO, nitric oxide; RBC, red blood cell.

with an electromagnetic shockwave source (Storz Medical, Kreuzlingen, Switzerland).

While shockwave therapy uses sound waves, it differs from ultrasound in frequency and application. Therefore, it falls outside the strict definition of ultrasound but remains “ultrasound-adjacent.” Free radical and epoxide production are theorized mechanisms of action.³ This may account for the immediate analgesic effects. Myocardial perfusion and function improvements may be the result of shockwave therapy-stimulated neo-angiogenesis.³ ECSWT is not yet indicated as a standard treatment in cases of **myocardial ischemia**, which is likely due to long duration of treatment, which requires transducer positioning through an intercostal space by a trained sonographer, despite potential upregulation of VEGF-Flt system.⁴⁷ Interestingly, Kikuchi et al. (2010) conducted a double-blind, placebo-controlled study of ECSWT in individuals with severe angina pectoris.⁴³ They found that ECSWT significantly improved symptoms and exercise tolerance compared to placebo, although a larger study is needed. Fukumoto et al.⁴⁴ (2006) conducted a study of ECSWT in 9 patients with end-stage coronary artery disease. They found that ECSWT improved myocardial perfusion in the ischemic area as assessed by dipyridamole stress thallium scintigraphy, and effects persisted for 12 months. ECSWT may be a particularly good option for patients with severe coronary artery disease who are not candidates for revascularization procedures.

Pulsed Cavitation Focused Ultrasound

The pulsed cavitation focused ultrasound is engineered to enhance the functionality of calcified aortic valves (AVs) through noninvasive ultrasound-based therapy. Its mechanism involves generating microscopic cavitation bubbles using short, high-pressure ultrasound pulses focused on calcified tissues, such as those seen in calcific AV stenosis. As these bubbles collapse, they create shock waves that mechanically soften the targeted calcified valve leaflets. In the study by Messas et al.⁴⁸ (2021), the researchers evaluated the feasibility and performance of noninvasive ultrasound-based therapy in ten patients with severe symptomatic AV stenosis. In this study, patients who received the most ultrasound-based therapy time and energy demonstrated encouraging results. More recently, Messas et al. demonstrated improved AV function up to 6 months after noninvasive ultrasound-based therapy in 40 adult patients with severe symptomatic AV stenosis.⁴⁹

Hypertension

There is early and growing evidence for its important role of HIFU in the treatment of **resistant arterial hypertension** through renal denervation.⁵⁰ A clinical study⁵⁰ demonstrated that noninvasive renal denervation using HIFU resulted in a mean reduction in 24-hour ambulatory blood pressure of -13.1/-7.6 (standard deviation [SD] 9.5/8.5) mmHg, -14.9/-9.0 (SD 8.7/7.0) mmHg, and -11.4/-4.8 (SD 4.8/4.8) mmHg from baseline to 1, 3, and 6 months, respectively.

Invasive Techniques

Thrombosis and Embolism

The EkoSonic Endovascular System (EKOS) catheter is marketed (Boston Scientific) as a Food and Drug Administration (FDA)-approved catheter-based treatment that uses a combination of ultrasound waves and drugs to dissolve blood clots that cause pulmonary embolism or deep vein thrombosis. Its clinical utility was shown to be effective in the ULTIMA,⁵¹ SEATTLE II,⁵² OPTALYSE,⁵³ and KNOCOUT⁵⁴ trials. In patients with pulmonary embolism at intermediate risk, the ULTIMA study⁵¹ (n = 59) found ultrasound-assisted catheter-directed thrombolysis to be superior to anticoagulants alone up to 24 hours postprocedure, with no substantial bleeding risk. In patients with acute massive and submassive pulmonary embolism, SEATTLE II trial⁵² (n = 150) found ultrasound-facilitated, catheter-directed, low-dose fibrinolysis decreased right ventricular dilation, pulmonary pressure, anatomic thrombus burden, and minimized intracranial hemorrhage up to 72 hours post-procedure. OPTALYSE trial⁵³ (n = 101) suggested that lower drug doses and shorter infusion times of ultrasound-facilitated catheter-directed thrombolysis were linked to improved right ventricular function and reduced clot burden, with low risk of bleeding up to 48 hours post-procedure. KNOCOUT trial⁵⁴ (n = 1500) furthered the understanding of OPTALYSE protocol adoption and showed a reduction of the right ventricular to left ventricular diameters (RV/LV) ratio up to 14 days post-procedure. Future large-scale, multicenter clinical trials are needed to determine the efficacy and safety of such protocols in the long term.

Shockwave Medical has FDA-approved devices utilizing intravascular lithotripsy for managing severely calcific plaque lesions that use acoustic shockwaves in a balloon-based delivery system.^{55,56} LIPUS is also

investigated for ultrasound-assisted thrombolysis of **acute pulmonary embolism**,^{57–59} right ventricular dysfunction, and heart failure with preserved left ventricular ejection fraction (in animal models^{60,61}). Previous studies compared the effectiveness of ultrasound-assisted catheter-directed thrombolysis and conventional catheter-directed thrombolysis and concluded that both were equally effective for the treatment of acute pulmonary embolism.^{58,59} However, it is not recommended as a standard treatment for acute pulmonary embolism. There was no clear advantage of using one treatment over the other; nevertheless, the rate of moderate or severe bleeding was higher in ultrasound-assisted thrombolysis intervention.⁵⁸ Similarly, a Dutch study⁵⁷ concluded that ultrasound-assisted thrombolysis is associated with a high rate of major bleeding in patients with high-risk pulmonary embolism. Therefore, caution needs to be taken particularly in individuals at risk of bleeding.

Arrhythmias

Ultrasound-based therapy benefits still outweigh the harms in **cardiac arrhythmias** (HIFU, moderate quality evidence^{16,62–64}). If medical therapy is not effective in treating cardiac arrhythmia, an ablation can be performed with the goal of thermally removing the arrhythmogenic tissue. This can be used to ablate circuits to stop arrhythmia re-entrant circuits or to inhibit areas of focus. HIFU using the epicardial method has been applied in cardiac surgeries. Reyes et al.⁶² (2016) reported the results of the National Spanish Registry of HIFU ablation for atrial fibrillation. The overall efficacy of atrial fibrillation ablation with the HIFU Epicor system was 63.9 and 45.9% at 24- and 36-month follow-up. The outcomes of a multicenter study of surgical epicardial HIFU ablation for atrial fibrillation were reported by Ninet et al. (2005).⁶³ Eighty percent of individuals with persistent atrial fibrillation associated with long-standing structural heart disease were successfully cured with epicardial, off-pump, beating-heart ablation using HIFU. The findings of a research on epicardial HIFU for the surgical management of atrial fibrillation were published in 2009 by Mitnovetski et al.⁶⁴ At the time of publication, 10/13 (77%) of the individuals had sinus rhythm; one needed a permanent pacemaker implantation; one was in atrial fibrillation; and another was in atrial flutter. Schopka et al. (2010) reported the results of a prospective clinical study of epicardial HIFU ablation for atrial fibrillation.¹⁶ Although 62% of patients at 6 months had sinus rhythms, overall conversion rates were lower than in earlier publications. Moreover, preclinical research indicates that an endocardial strategy for treating mid-myocardial or epicardial ventricular arrhythmias may be effective with HIFU catheter ablation.⁸

Hypertension

There is early and growing evidence for its important role of HIFU in the treatment of **resistant arterial hypertension** through renal denervation.⁶⁵ The sympathetic nervous system in the kidneys plays a crucial role in regulating and contributing to the development of hypertension. Consequently, ultrasound-based renal denervation holds the potential to reduce blood pressure. This technique is usually performed with an endovascular approach. The recent RADIANCE II randomized controlled trial demonstrated efficacy and safety of ultrasound renal denervation in reducing blood pressure in patients with systemic hypertension without postoperative major adverse events.⁶⁶ This resulted in FDA approval of PARADISE ultrasound denervation for treatment of hypertension in 2023.

Ischemic Heart Failure

Recently, an Austrian trial indicated that direct cardiac shockwave therapy, combined with coronary bypass surgery, improves left ventricular ejection fraction (LVEF) and physical capacity in patients with ischemic heart failure. Larger trials are needed to confirm if direct

cardiac shockwave therapy improves myocardial function, as suggested here.⁶⁷

Miscellaneous

The role of ultrasound-based therapy is also being explored for managing **incompetent veins** (noninvasively and thermogenically closing a malfunctioning vein permanently^{68,69}), **hypertrophic cardiomyopathy**, and **external cardiac pacing** through cardiac stimulation (preclinical data^{70–72}). A case report⁶⁸ demonstrated that extracorporeal HIFU treatment of an incompetent perforating vein in a 95-year-old man with active venous ulcers was successful. Three months after HIFU, reflux was eliminated, and the two ulcers that were initially active had healed. Whiteley⁶⁹ also suggested that HIFU is a promising treatment for varicose veins and venous leg ulcers. The effectiveness of therapeutic ultrasonography, at either high or low frequency, in accelerating the healing of venous leg ulcers is unknown.⁷³ There is some limited evidence of enhanced healing when using therapeutic ultrasound⁷³; however, further research is needed to evaluate the long-term efficacy and safety of HIFU in these patients.

General Considerations and Implications

Ultrasound-based therapies represent emerging modalities in the management of complex cardiovascular conditions. As these innovative approaches gain traction, several key considerations must be thoroughly addressed to ensure their effective integration into clinical practice.

Safety and Efficacy

The foremost concern with the introduction of any new therapeutic technology is its safety and efficacy. HIFU procedures require vigilance due to possible thermal injury, potentially causing unintended tissue damage if not properly controlled and monitored. Cooling and temperature monitoring are crucial for mitigating this risk. ECWT and ultrasound-facilitated thrombolysis can induce mechanical stress, leading to vascular injury if not carefully administered. Careful selection of acoustic parameters and cavitation monitoring are essential for safety.^{6,74,75} Ultrasound-based therapies must be validated through rigorous, large-scale, randomized clinical trials that not only assess the immediate therapeutic outcomes but also long-term effects on patient health and disease progression. For cardiologists, involvement goes beyond clinical application; it extends to active participation in research and critical evaluation of both positive outcomes and potential adverse effects. Ensuring that these technologies do more good than harm is paramount, and this can only be achieved with comprehensive data that supports their use in specific cardiovascular scenarios.

Integration Into Practice

Integrating therapeutic ultrasound into routine cardiology practice requires a multifaceted approach. Firstly, practitioners need detailed training to understand the nuances of ultrasound technologies, including the technical aspects of equipment and the physiological implications of their use. Secondly, clear clinical guidelines must be developed to delineate indications for use, procedural protocols, and contraindications. These guidelines should be evidence-based to ensure standardized care across different health care settings. Furthermore, effective integration relies on interdisciplinary collaboration among cardiologists, sonographers, bioengineers, and other health care professionals to fine-tune the application of these technologies and tailor them to individual patient needs.

Patient Selection

The success of ultrasound-based therapy greatly depends on selecting the right patients. Factors such as the specific cardiac anatomy, the nature and stage of the CVDs, and the overall health condition of the patient play critical roles in determining the appropriateness of ultrasound-based therapies. For instance, patients with complex cardiac geometries or those with certain types of tissue characteristics may benefit more from HIFU or ESWT, whereas others may be better suited for LIPUS based on the desired therapeutic outcomes and risk profiles. Precision in patient selection ensures that the applied technology provides maximum benefit and minimizes risks, thus leveraging the full potential of therapeutic ultrasound in personalized medicine.

Cost-Effectiveness and Accessibility

Another significant consideration is the cost-effectiveness of these ultrasound technologies compared to traditional treatments. Health economics should be considered, assessing not only the direct costs associated with the technology itself but also the potential reduction in overall health care expenses through improved management of CVDs. Additionally, the accessibility of these technologies in different health care settings, including under-resourced environments, must be evaluated to ensure equitable health care delivery.

Future Perspectives

Looking forward, the continued advancement in ultrasound technology promises to expand the boundaries of what is currently achievable in cardiovascular therapies. Innovations in ultrasound technology could lead to more precise interventions, reduced invasiveness, and enhanced patient outcomes. Embracing these technologies also means preparing for a future where cardiology practice may increasingly rely on advanced imaging and therapeutic modalities as integral components of cardiovascular care.

Conclusion

The efficacy of therapeutic ultrasound in ST elevation myocardial infarction and cardiac arrhythmia has been demonstrated through clinical trials when used alongside standard therapy. However, its potential for treating tumors, resistant arterial hypertension, venous leg ulcers, and acute pulmonary embolism is not yet fully established. There are concerns regarding potential side effects, such as excessive bleeding in individuals at risk. The ongoing challenge lies in determining the optimal intensity, frequency, and duration of ultrasound exposure for patients with CVDs.

CRediT Authorship Contributions

Amit Bansal drafted the original manuscript and submitted it. Later, additional authors were invited to contribute. Bernard Bulwer and Ricarda von Krüchten then undertook extensive rewriting of the manuscript. Jagkirat Singh provided a review of the revised version. Rajan Rehan created the tables and reviewed the overall work. Finally, Ata Doost also offered extensive feedback and contributed to the generation of the second figure.

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